

MODULE 24

SQL Real World Projects

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

SQL Real World Projects

E-Commerce Schema

```
products(id, name, price, stock)
orders(id, customer_id, total, status)
order_items(order_id, product_id, qty)

-- Top selling products
SELECT p.name, SUM(oi.qty) AS sold
FROM products p
JOIN order_items oi ON p.id = oi.product_id
GROUP BY p.id ORDER BY sold DESC LIMIT 10;
```

HR Hierarchy

```
WITH RECURSIVE tree AS (
SELECT emp_id, name, manager_id, 0 AS level
FROM employees WHERE manager_id IS NULL
UNION ALL
SELECT e.emp_id, e.name, e.manager_id, t.level+1
FROM employees e JOIN tree t ON e.manager_id = t.emp_id
)
SELECT REPEAT(' ', level), name FROM tree;
```

School Analytics

```
SELECT d.dept_name,
COUNT(*) AS students,
ROUND(AVG(gpa), 2) AS avg_gpa,
COUNT(CASE WHEN gpa >= 3.5 THEN 1 END) AS honor_roll
FROM students s
JOIN departments d ON s.dept_id = d.dept_id
GROUP BY d.dept_id ORDER BY avg_gpa DESC;
```

Inventory Alert

```
SELECT name, stock,
CASE
WHEN stock = 0 THEN 'OUT OF STOCK'
WHEN stock < reorder_level THEN 'REORDER NOW'
ELSE 'OK'
END AS status
FROM products
WHERE stock <= reorder_level;
```

Interview Q and A

1. DELETE vs TRUNCATE vs DROP?

DELETE: rollback-able, triggers fire

TRUNCATE: fast, no rollback, no triggers

DROP: removes table entirely

2. HAVING vs WHERE?

WHERE: filters rows before GROUP BY

HAVING: filters groups after GROUP BY

3. Clustered vs Non-clustered index?

Clustered: data physically sorted by key

Non-clustered: separate pointer structure